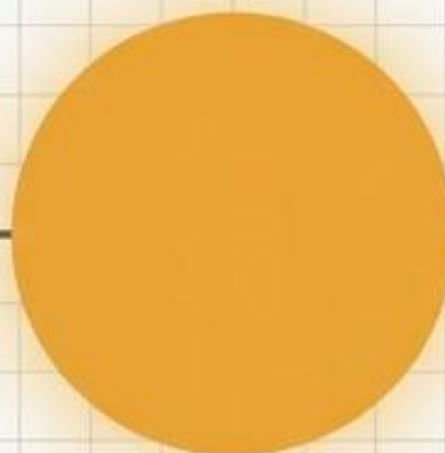


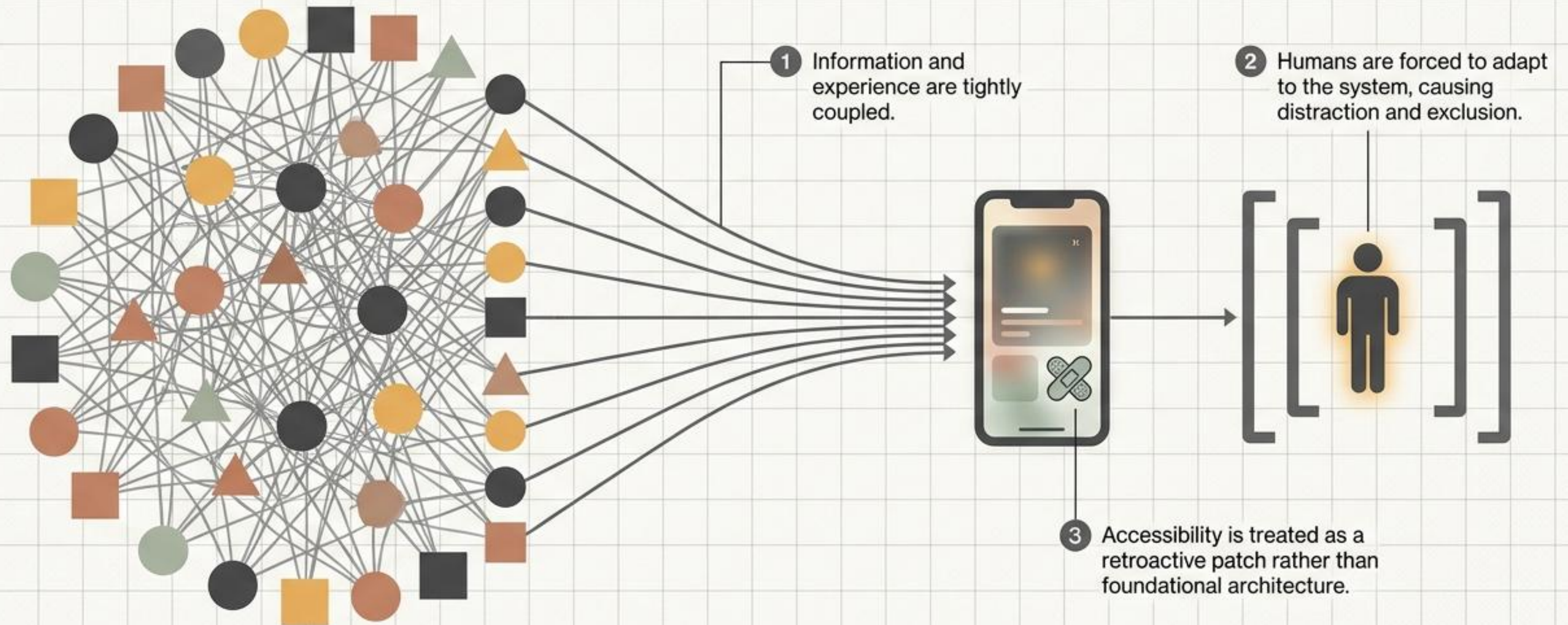
HUMAN FIRST. CONTEXT APPROPRIATE.

ADE Human-Machine Framework

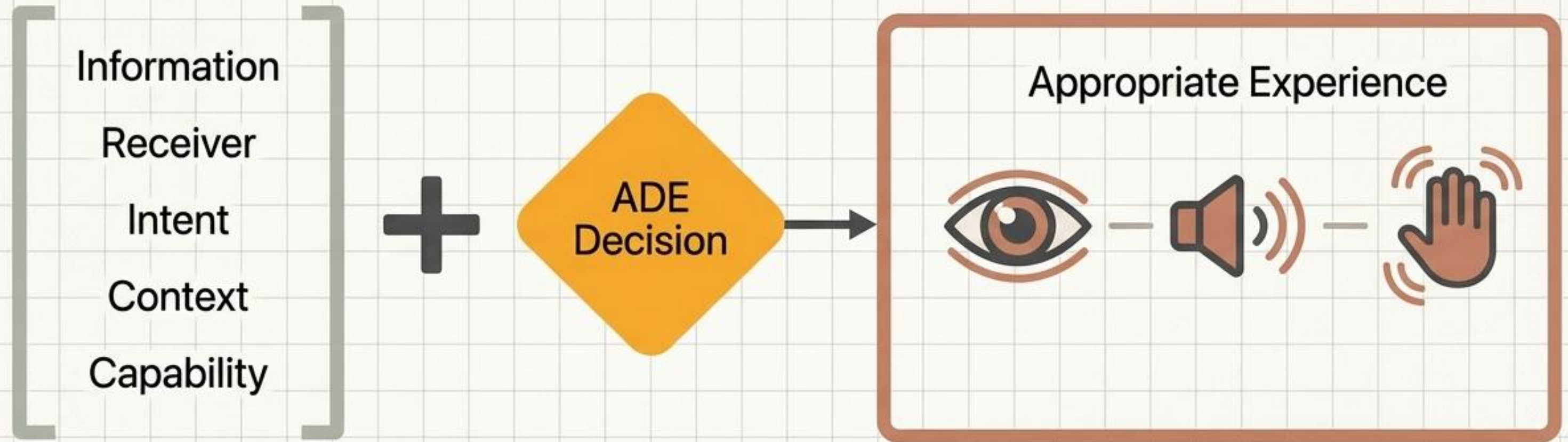
Architectural Hypothesis v0.1



Information is built for screens first, humans second.



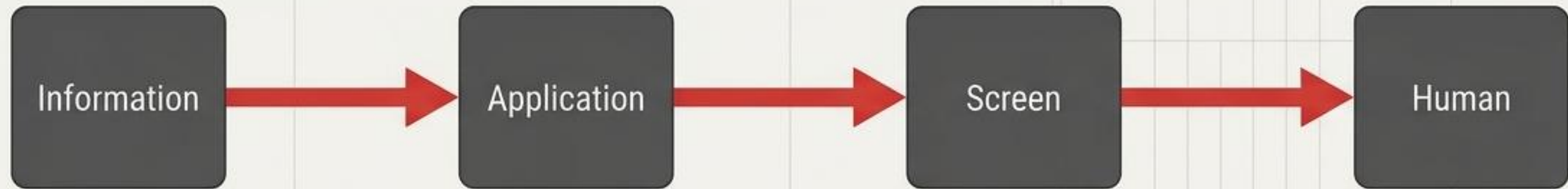
Shared Meaning, Independent Experience.



The source communicates meaning. The receiving environment determines the experience.
Information describes what matters; context determines how it is delivered.

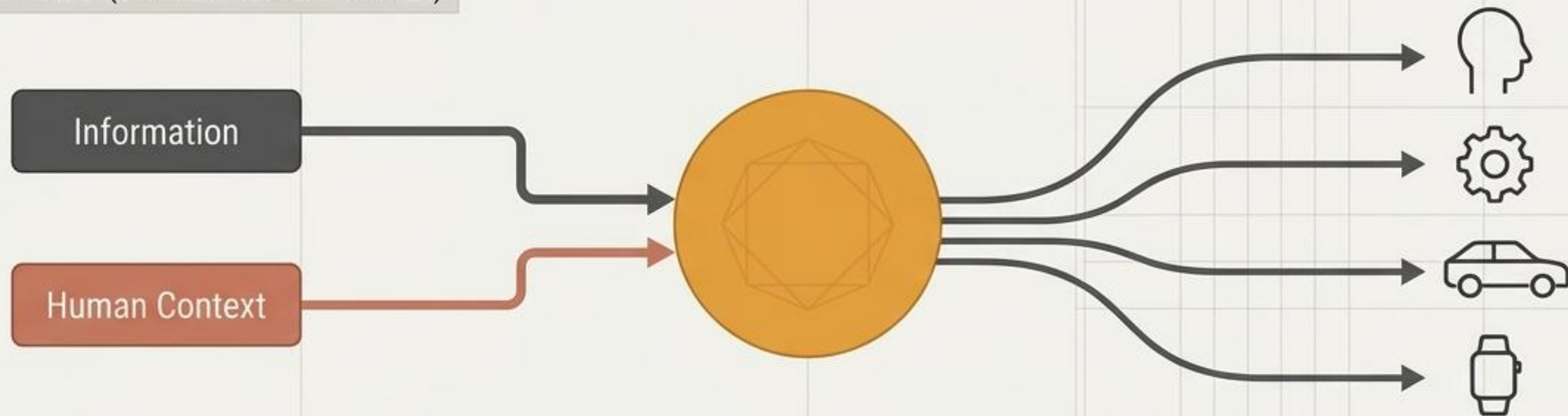
Decoupling the Information Architecture

CURRENT MODEL (RIGID & LINEAR)



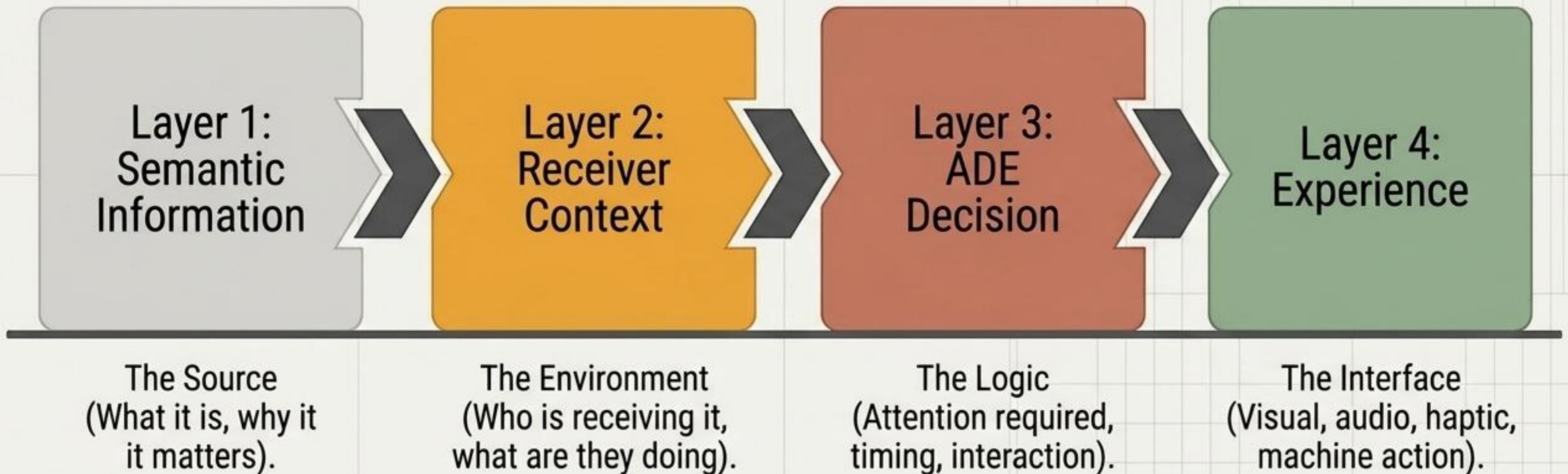
The original source must know the final receiver and dictate the presentation model.

ADE HYPOTHESIS (DYNAMIC ROUTING MATRIX)



One semantic source routes dynamically to humans, machines, and future systems not yet invented.

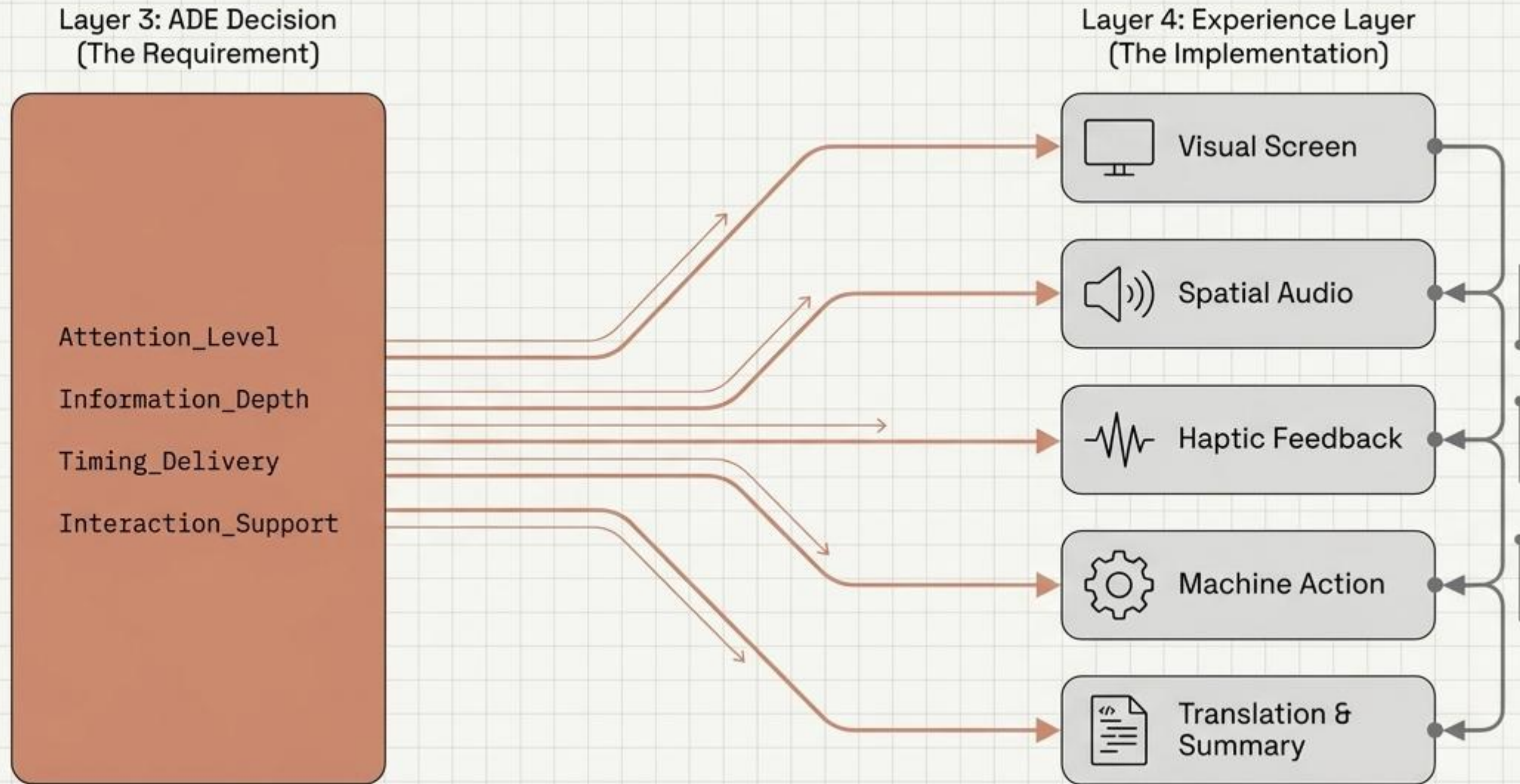
The 4-Layer Routing Architecture



The Inputs: Meaning Meets Context

Semantic Information (The Source)		Receiver Context (The Environment)	
[WHAT]	Entity, event, action, state.	[RECEIVER]	Human, machine, system.
[WHERE / WHEN]	Location, scope, duration.	[INTENT]	Purpose or objective of the receiver.
[SIGNIFICANCE]	Urgency, consequence, confidence.	[ACTIVITY]	Current physical or cognitive activity.
[REQUIREMENT]	Action or response to be taken.	[CAPABILITY]	Available sensory or interface options.
[APPLICABILITY]	Who or what is affected.		

The Output: Defining Requirements, Not Interfaces



ADE calculates the precise human requirement, and hands off to the local ecosystem to fulfill the interface.

In Practice: One Source, Many Experiences

SEMANTIC PAYLOAD

Event:
Wildfire
Approaching.

Significance:
Critical.

Requirement:
Seek Shelter.



	Driver in Vehicle	Person at Home	Irrigation System
Context Data	Attention limited, hands busy.	Attention available, screens present.	Machine receiver, no human.
ADE Decision	Immediate timing, Audio format.	Immediate timing, Visual + Audio format.	Immediate timing, Machine logic format.
Final Experience	Spoken alert over car speakers. "Evacuate now."	Flashing smart lights and TV screen takeover.	Sprinklers activate automatically to wet the roof.

Defining the Scope Boundary

What it is NOT

- ✗ Not a screen replacement (screens remain vital when appropriate).
- ✗ Not solely an accessibility standard (though disability access benefits immensely).
- ✗ Not an attempt to replace or reinvent existing data standards.

What it IS

- ✓ A universal human-first routing architecture.
- ✓ Beneficial for drivers, workers, noisy environments, and cross-language communication.
- ✓ A missing integration layer to synthesize human intent and environmental capability.

Building Upon the Existing Ecosystem

The
Interface

Accessibility UI:
WCAG, WAI-ARIA, WAI-Adapt

Machine/Device:
W3C Web of Things (WoT), OPC-UA

The
Missing
Link

ADE Framework:
The missing decision logic between
semantic data and presentation standards.

Semantic Data:
RDF, JSON-LD,
Schema.org

Context & Env:
NGSI-LD

Emergency:
Common Alerting
Protocol (CAP)

The Foundation

Open Research Questions

[Q1]

What is the minimum semantic information required to support appropriate contextual interpretation?

[Q2]

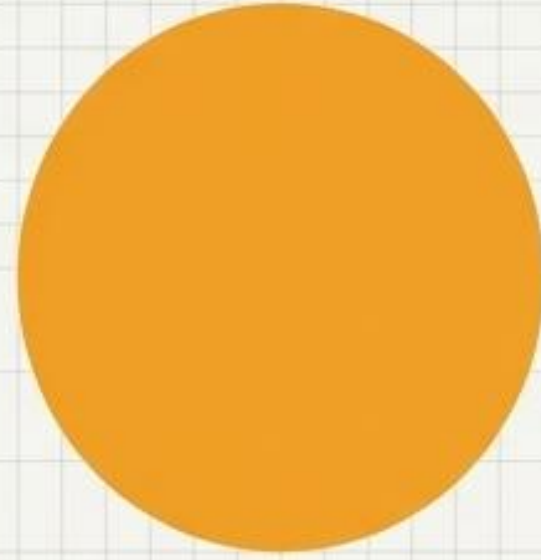
Can a single common decision model support both human and machine receivers simultaneously?

[Q3]

Can information meaning truly remain stable while experiences vary across infinite edge-case contexts?

[Q4]

Which ADE concepts represent genuine architectural gaps versus existing but underutilized standards?



Tell me where I am wrong.

This is a v0.1 architectural hypothesis. Tear it apart, stress-test the edges, and help build the human-first internet.

`github.com/ADE-Human-Machine-Framework`